How CMFGEN Works — and the Non-Monotonic Velocity Solver

A simple explainer for first-year PhD students

The big picture

CMFGEN (CMF GENeral) is a code that computes the spectrum of light coming out of an expanding atmosphere — a supernova, a stellar wind, or any outflowing envelope. You give it the structure of the gas at one instant in time — the radius, density, velocity, temperature and chemical composition at each point — and it tells you:

- what the emergent spectrum looks like, and
- what the ionisation and excitation state of every atom is at every depth.

The hard part is that the radiation field and the atomic populations depend on *each other*. The radiation field determines which atoms are ionised and excited; the ionisation and excitation determine the opacity and emissivity, which in turn determine the radiation field. So CMFGEN has to solve a coupled problem self-consistently. It does this by *iteration*.

1 Radiative transfer terms: a glossary

The rest of this document uses a standard set of radiative-transfer quantities. This section defines each one, states its physical meaning, and notes the units and sign conventions used inside CMFGEN. Readers already comfortable with the notation can skip to §2.

1.1 Specific intensity I_ν

The **specific intensity** (or *specific radiative intensity*, sometimes called *brightness*) is the most fundamental quantity in radiative transfer. It is the energy flowing in a given direction, per unit area, per unit time, per unit frequency, per unit solid angle:

$$I_\nu(\mathbf{r}, \hat{\mathbf{n}}, t) \;=\; \frac{dE}{dA_\perp \, dt \, d\nu \, d\Omega}, \tag{1}$$

with SI units $\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}\,sr^{-1}}$. I_ν depends on position $\mathbf{r}$, direction $\hat{\mathbf{n}}$, frequency ν and time t. In CMFGEN the time dependence is frozen out (a snapshot), and the geometry is spherical, so the direction is parametrised by the ray cosine μ (see below) and the impact parameter p (see below).

The intensity is what the formal solver integrates along each ray. It is also the quantity that the non-monotonic solver must keep finite and non-negative: a negative or NaN I_ν propagates through every moment and every rate equation.

1.2 Ray cosine μ and impact parameter p

In spherical geometry a ray is labelled by its **impact parameter** p, the closest approach distance to the centre, and by the local **ray cosine**

$$\mu \;=\; \hat{\mathbf{n}} \cdot \hat{\mathbf{r}} \;=\; \cos\theta, \tag{2}$$

where θ is the angle between the ray direction and the outward radial direction. Along a given ray, μ varies with radius as $\mu = \sqrt{1 - p^2/r^2}$; $\mu = +1$ is an outward radial ray, $\mu = -1$ is an inward radial ray, and $\mu = 0$ is a tangential ray at $p = r$.

CMFGEN discretises the rays by a set of impact parameters $p_1, \ldots, p_{N_P}$ (typically a Gauss quadrature on $[0, R_{\max}]$) and integrates the transfer equation along each one. The angle quadrature weights turn the ray intensities into the angular moments.

1.3 Opacity χ_ν

The **opacity** (or *extinction coefficient*) is the probability per unit path length that a photon of frequency ν is removed from the beam (by absorption or scattering out of the direction/frequency bin). It has units of cm^{-1} and is related to the cross section σ_ν and number density n by $\chi_\nu = n\,\sigma_\nu$ (summed over all species). In CMFGEN χ_ν includes:

- bound–bound (line) opacity;
- bound–free (photoionisation) opacity;
- free–free (bremsstrahlung) opacity;
- electron (Thomson) scattering opacity.

The opacity is set by the atomic populations, which is why it couples back to the rate equations.

1.4 Emissivity η_ν

The **emissivity** is the energy added to the beam per unit volume, time, frequency and solid angle: $[\eta_\nu] = \mathrm{erg\,s^{-1}\,cm^{-3}\,Hz^{-1}\,sr^{-1}}$. It includes thermal (free–free, free–bound) emission, bound–bound (spontaneous + stimulated) emission, and scattering into the beam. In the transfer equation η_ν is the source term and $\chi_\nu I_\nu$ is the sink term.

1.5 Source function S_ν

The **source function** is the ratio of emissivity to opacity,

$$S_\nu \;=\; \frac{\eta_\nu}{\chi_\nu}, \tag{3}$$

with the same units as I_ν. In local thermodynamic equilibrium (LTE) $S_\nu = B_\nu(T)$, the Planck function. In non-LTE (the general CMFGEN case) S_ν differs from B_ν because the populations are not at their LTE values and because scattering contributes to η_ν and χ_ν without being tied to the local T.

In the comoving-frame formal solver CMFGEN works with a **generalised source function** $S' = S + Q\,I_{\mathrm{upwind}}$ (see §2), which absorbs the velocity-coupling contribution into an effective source.

1.6 Optical depth τ_ν

The **optical depth** along a path is the dimensionless integral of the opacity,

$$\tau_\nu \;=\; \int \chi_\nu \, ds. \tag{4}$$

A photon has probability $e^{-\tau_\nu}$ of traversing the path without interaction. CMFGEN also uses a **generalised optical depth** $\hat{\tau} = \hat{\chi}\,ds$, where $\hat{\chi}$ is the generalised opacity that includes the velocity-coupling term (see below). The non-monotonic solver exists precisely because $\hat{\chi}$ can go negative in the standard formulation when the velocity field is non-monotonic.

1.7 Mean intensity J_ν

The **mean intensity** is the angle average of I_ν,

$$J_\nu \;=\; \frac{1}{4\pi}\oint I_\nu \, d\Omega \;=\; \frac{1}{2}\int_{-1}^{+1} I_\nu(\mu)\, d\mu, \tag{5}$$

with the same units as I_ν. J_ν is the quantity that enters the radiative rates in the statistical-equilibrium equations: a bound–bound rate is proportional to the profile-averaged $\bar{J}$ (see below), and a photoionisation rate is proportional to J_ν integrated over the ionisation edge. Computing J_ν from the ray intensities is the main job of the formal solver, and it is the quantity that the non-monotonic solver replaces.

1.8 First moment H_ν (the flux)

The **first moment** (or *Eddington flux*) is the angle-weighted average,

$$H_\nu \;=\; \frac{1}{4\pi}\oint \mu\, I_\nu \, d\Omega \;=\; \frac{1}{2}\int_{-1}^{+1} \mu\, I_\nu(\mu)\, d\mu. \tag{6}$$

$4\pi H_\nu$ is the *net flux* (energy per unit area per unit time per unit frequency) through a spherical shell. The luminosity is $L_\nu = 16\pi^2 r^2 H_\nu$. At a velocity reversal H_ν passes through zero (inward and outward flux cancel), which is why the non-monotonic configuration requires `N_ON_J` rather than `G_ONLY`: the ratio N/H would be singular.

1.9 Second moment K_ν

The **second moment** is

$$K_\nu \;=\; \frac{1}{4\pi}\oint \mu^2\, I_\nu \, d\Omega \;=\; \frac{1}{2}\int_{-1}^{+1} \mu^2\, I_\nu(\mu)\, d\mu. \tag{7}$$

K_ν measures the radiation pressure ($P_{\rm rad} = (4\pi/c)K_\nu$) and the degree of anisotropy. The ratio

$$f_{\rm Edd} \;=\; \frac{K_\nu}{J_\nu} \tag{8}$$

is the **Eddington factor**, which closes the moment hierarchy. In the diffusion limit $f_{\rm Edd} \to 1/3$ (isotropic); in the free-streaming limit $f_{\rm Edd} \to 1$ (radial). CMFGEN computes $f_{\rm Edd}$ from the ray quadrature and feeds it into the moment equations (or, in the non-monotonic path, into the BA matrix).

1.10 Third moment N_ν

The **third moment** is

$$N_\nu \;=\; \frac{1}{4\pi}\oint \mu^3\, I_\nu \, d\Omega \;=\; \frac{1}{2}\int_{-1}^{+1} \mu^3\, I_\nu(\mu)\, d\mu. \tag{9}$$

It appears in the variable-Eddington-factor closure and in the comoving-frame moment equations (the term that couples H to K and N). The ratio N_ν/J_ν is used in the `N_ON_J` closure that the non-monotonic solver requires.

1.11 Profile-averaged mean intensity $\bar{J}$

For a bound–bound transition $i \to j$ the radiative rate is not proportional to J_ν at a single frequency but to the mean intensity *averaged over the line profile* $\phi(\nu)$:

$$\bar{J}_{ij} \;=\; \int_0^\infty J_\nu \, \phi_{ij}(\nu)\, d\nu. \tag{10}$$

In the Sobolev / large-velocity-gradient limit the profile is a delta function at the resonance frequency, so $\bar{J}$ reduces to J at the single resonance point. In the comoving-frame formal solution the profile is resolved on the frequency grid, and $\bar{J}$ is computed by integrating J_ν over the line. The Sobolev routines (`SOBEW`, `SOBJBAR`, etc.) compute $\bar{J}$ directly from the Sobolev optical depth and escape probability; these are the routines that the Dessart+2015 sigma guard stabilises for non-monotonic flow.

1.12 Velocity-gradient parameter σ

The **velocity-gradient parameter** is the dimensionless logarithmic gradient of the expansion velocity,

$$\sigma \;=\; \frac{d\ln V}{d\ln R} - 1 \;=\; \frac{R}{V}\frac{dV}{dR} - 1. \tag{11}$$

For homologous expansion ($V \propto R$) $\sigma = 0$. For a decelerating shell ($dV/dR < 0$) $\sigma < -1$, which is the regime that breaks the standard solver. CMFGEN maintains two arrays: `SIGMA_TRUE` (the unclamped physical value, used by the non-monotonic formal solver) and `SIGMA` (clamped to ≥ -0.999, used by every other routine for stability).

1.13 Sobolev optical-depth scale Γ

Γ is the comoving-frame frequency-gradient scale that appears in the generalised opacity,

$$\Gamma \;=\; \frac{V}{R}\big(1 + \sigma\,\mu^2\big), \tag{12}$$

with units of s^{-1}. It measures how fast a photon's comoving frequency shifts per unit path length. In monotonic flow $\Gamma > 0$ (photons redshift outward); in compression zones $\Gamma < 0$ (photons blueshift as they move into the compression). The sign of Γ is what the non-monotonic solver uses to choose the upwind frequency.

1.14 Generalised opacity $\hat{\chi}$

In the comoving frame the transfer equation has an extra term from the frequency shift. CMFGEN absorbs it into a **generalised opacity**,

$$\hat{\chi} \;=\; \chi \;+\; \frac{\Gamma}{\Delta \log \nu}, \tag{13}$$

where $\Delta \log \nu$ is the frequency-grid spacing. In the standard (monotonic) solver $\Gamma > 0$ so $\hat{\chi} > \chi > 0$. In non-monotonic flow Γ can be negative enough to make $\hat{\chi} < 0$, which is unphysical. The Knop et al. (2009) fix replaces Γ with $|\Gamma|$ in the opacity (so $\hat{\chi} > 0$ always) and carries the sign in the source function instead.

1.15 Summary table

Symbol	Name	Units
I_ν	specific intensity	$\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}\,sr^{-1}}$
J_ν	mean intensity (0th moment)	same as I_ν
H_ν	first moment (flux)	same as I_ν
K_ν	second moment (rad. pressure)	same as I_ν
N_ν	third moment	same as I_ν
$\bar{J}$	profile-averaged J	same as I_ν
η_ν	emissivity	$\mathrm{erg\,s^{-1}\,cm^{-3}\,Hz^{-1}\,sr^{-1}}$
χ_ν	opacity	$\mathrm{cm^{-1}}$
S_ν	source function ($=\eta/\chi$)	same as I_ν
τ_ν	optical depth	dimensionless
$\hat{\chi}$	generalised opacity	$\mathrm{cm^{-1}}$
$\hat{\tau}$	generalised optical depth	dimensionless
μ	ray cosine	dimensionless
p	impact parameter	cm
σ	velocity-gradient parameter	dimensionless
Γ	Sobolev frequency scale	$\mathrm{s^{-1}}$
$f_{\rm Edd}$	Eddington factor (K/J)	dimensionless

2 The two equations CMFGEN solves

2.1 1. Radiative transfer (the formal solution)

At every point in the ejecta, the specific intensity I_ν obeys the radiative transfer equation. In the **comoving frame** (the frame moving with the gas) this is, schematically,

$$\frac{dI_\nu}{ds} \;=\; \eta_\nu \;-\; \chi_\nu\, I_\nu \;+\; \text{velocity-coupling terms}, \tag{14}$$

where η_ν is the emissivity, χ_ν the opacity, and s the path length along a ray. The "velocity-coupling terms" appear because we are in an *expanding* medium: in the comoving frame, the frequency of a photon *shifts* as it propagates. In a monotonic outflow photons are redshifted as they travel outward, coupling each frequency to the next-lower frequency. This coupling is the key physical effect that distinguishes comoving-frame (CMF) radiative transfer from static radiative transfer.

CMFGEN's standard formal solver absorbs the velocity coupling into a **generalised opacity**,

$$\hat{\chi} \;=\; \chi \;+\; \frac{\Gamma}{\Delta \log \nu}, \qquad \Gamma \;=\; \frac{V}{R}\big(1+\sigma\,\mu^2\big), \qquad \sigma \;=\; \frac{d\ln V}{d\ln R} - 1, \tag{15}$$

where $\Delta \log \nu$ is the frequency-grid spacing, μ the ray cosine, and σ the *velocity-gradient parameter*. The transfer equation then becomes a standard short-characteristics problem,

$$\frac{dI}{d\hat{\tau}} \;=\; I - S', \qquad \hat{\tau} = \hat{\chi}\,dz, \qquad S' = S + Q\,I_{\rm upwind}, \qquad Q = \frac{\Gamma/\Delta\log\nu}{\hat{\chi}}, \tag{16}$$

with $S = \eta/\hat{\chi}$ the generalised source function and $I_{\rm upwind}$ the intensity at the upwind frequency (already computed in the decreasing-frequency sweep). The formal solution is performed for all rays (impact parameters p) and all frequencies, producing the moments J_ν, H_ν, K_ν of the radiation field.

2.2 2. Statistical equilibrium (rate equations)

For every atom and every level, the population n_i must satisfy "number in = number out":

$$\sum_{j \neq i} n_j R_{ji} \; - \; n_i \sum_{j \neq i} R_{ij} \; = \; 0. \tag{17}$$

The rates R_{ij} include:

- **Radiative rates**: $R_{ij}^{\rm rad} \propto \bar{J}_{ij}$, the mean intensity averaged over the line/continuum profile. These depend on the radiation field.
- **Collisional rates**: $R_{ij}^{\rm coll} \propto n_e$, depending on electron density and temperature.

So the populations depend on $\bar{J}$ (radiation), and $\bar{J}$ depends on the populations (through η_ν, χ_ν). That is the coupling.

3 The iteration scheme

CMFGEN does not solve both equations simultaneously. It alternates:

1. **Given current populations**, solve the radiative transfer equation to get the radiation field J_ν (and the Eddington factors, luminosity, etc.).
2. **Given** J_ν, solve the rate equations to get updated populations.
3. Repeat until the populations stop changing.

This is called **Λ-iteration** when step 1 uses a simple formal solution. Plain Λ-iteration converges very slowly, so CMFGEN uses several accelerations:

NG acceleration (Ng 1974): every few iterations, extrapolate the population sequence to try to jump ahead to the converged answer. Works well when the last few iterations are converging smoothly.

BA iterations (the "Ba" matrix, an approximate-Newton / Broyden-like correction): instead of just feeding J_ν back into the rate equations, linearise the *coupled* transfer + rate system and solve for a correction to the populations that accounts for how the radiation field will respond. This is much faster than pure Λ-iteration but more expensive per step and can be unstable.

A typical CMFGEN run cycles through blocks like

$$\text{LAMBDA} \; \rightarrow \; \text{LAMBDA} \; \rightarrow \; \text{NG} \; \rightarrow \; \text{FULL (BA)} \; \rightarrow \; \text{LAMBDA} \; \rightarrow \; \ldots$$

starting with cheap Λ iterations to get close, then switching to the more expensive BA/Full iterations to finish converging. The iteration strategy is controlled by the `IT_SPECIFIER` block in the input, which alternates between `LAMBDA(FIXT)`, `FULL(FIXT)`, `FULL(FIXBA)`, and `DO_NG` commands.

3.0.1 Current iteration strategy for SN2024pba

The SN2024pba production configuration uses the following iteration strategy, controlled by the `single_final` profile in `prepare_VADAT.py`:

1. **Cold start from LTE** (`LTE_EST=T`): initial level populations are set to LTE values computed from the hydrodynamic temperature and electron density. This provides a stable starting point that prevents the extreme population excursions that caused the earlier blow-up without `LTE_EST` (see §9).
2. **Pure Lambda iterations** (`DO_LAM_IT=T`): the first iterations are pure Lambda iterations with `MAX_LAM`=10^5, allowing large fractional population changes per iteration. This is stable with the Knop formal solver because $\hat{\chi} > 0$ always, so J_ν remains finite even when the populations are far from converged.

3. **Auto-switch to BA** (`DO_LAM_AUTO=T`): once the maximum population correction falls below 50%, CMFGEN automatically switches from Lambda iterations to full BA (linearisation) iterations. The BA solve couples the radiation field to the population correction, giving much faster convergence than pure Lambda iteration.
4. **Ng acceleration** (`DO_NG=T`, `IBEG_NG=50`, `BW_NG=60`, `ITS/NG=8`): Ng acceleration is applied every 8 iterations once at least 50 iterations have elapsed and the maximum correction is below `BEG_NG`=10%. The Ng implementation includes per-depth rejection (if any population changes by more than 10.1× or less than 0.09×), global cancellation (if more than 3 depths fail), and a post-Ng Lambda fallback. Ng is essential for the DIAG solver, which converges geometrically and would stall at $\sim 10^{10}$ without acceleration.
5. **Trace-species freezing** (`FIX_IMP=T`): trace species (low-abundance ions) are frozen in the BA matrix via `FIXPOP_IN_BA_V3`, preventing singular BA matrix columns from trace species with near-zero populations. This is particularly important for Fe IV and Fe V at late-time snapshots where their populations are set by photoionisation and can be extremely small.
6. **Temperature bootstrap** (`FIX_T=F`, `FIX_T_AUTO=T`): temperature is initially held fixed to the hydrodynamic value (via the `FIX_T_AUTO` bootstrap) while populations converge from LTE, then automatically released so CMFGEN solves for temperature self-consistently. This follows the Dessart et al. (2015) approach and prevents temperature oscillations during the cold-start phase.

The convergence sequence is typically:

LTE → Lambda → Lambda → … → BA (auto) → Ng → BA → Ng → … → converged

with the Knop formal solver handling the compression zones at every iteration so J_ν stays physical throughout.

3.1 The moment equations and the BA matrix

Instead of solving for J_ν directly from the ray quadrature, CMFGEN *optionally* solves the 0th and 1st moments of the transfer equation as a tridiagonal system. This couples adjacent depth points and gives a more stable J_ν than pure ray quadrature, especially in optically thick regions. The moment equations require Eddington factors (K/J and N/J), computed from the ray moments.

The BA (Band Acceleration) method computes the Jacobian of J_ν with respect to the populations ($dJ/d\chi$, $dJ/d\eta$) by linearising the moment equations. The resulting banded matrix couples adjacent depth points; it is solved for a population correction that accounts for the *global* sensitivity of J_ν to population changes — which is why BA converges much faster than pure Λ iteration.

4 Why the velocity matters: the Sobolev / monotonic assumption

In a standard supernova outflow, the velocity increases monotonically outward: $V(r)$ increases with r, so $dV/dR > 0$ and $\sigma = d\ln V/d\ln R - 1 > -1$. This is the "Hubble flow" regime, and it is hugely convenient because of the **Sobolev approximation**: a photon at frequency ν only interacts with material where the Doppler shift brings it into resonance, and if $dV/dR > 0$ that resonance point is *unique*. Each frequency maps to exactly one depth. The transfer problem becomes local and tractable.

Even when CMFGEN does a full formal solution (not the pure Sobolev approximation), most of its machinery — the characteristic rays, the frequency-to-depth mapping, the way the source function is built up — *assumes* that $V(r)$ is monotonic increasing outward. The characteristics don't cross, and the generalised opacity $\hat{\chi} = \chi + \Gamma/\Delta\log\nu$ is always positive because $\Gamma > 0$.

5 The problem: non-monotonic velocity

In interacting supernova systems — SNe IIn, SNe with circumstellar interaction, shock breakout, cold dense shells, reverse shocks — the velocity field can be **non-monotonic**:

- **Shocks**: V jumps discontinuously at the shock front.
- **Velocity reversals**: V decreases with R in the post-shock region.
- **Cold dense shells**: ejecta decelerates abruptly, $V(R)$ has a local minimum.
- **Reverse shocks**: inflowing material ($V < 0$) behind a forward shock.

In these cases $\sigma = d\ln V/d\ln R - 1$ can be < -1, meaning the comoving-frame frequency gradient $\Gamma = (V/R)(1 + \sigma\mu^2)$ can be *negative* for some ray angles.

5.1 Why the standard solver fails

The standard CMFGEN formal solver was designed for monotonic outflows where $\Gamma > 0$ always. When $\Gamma < 0$ (non-monotonic flow), the generalised opacity

$$\hat{\chi} \;=\; \chi \;+\; \frac{\Gamma}{\Delta \log \nu}$$

can become *negative*. A negative opacity is unphysical and causes a cascade of failures:

- **Negative optical depths** ($\hat{\tau} < 0$).
- **Exponentially growing intensities** ($e^{+\hat{\tau}}$ instead of $e^{-\hat{\tau}}$).
- **NaN propagation** through the ray integration.
- **Instability** in the moment equations.
- **BA matrix singularity**.

5.2 The infrastructure clamp (and why it is not enough)

To prevent outright crashes, CMFGEN clamps σ everywhere:

$$\texttt{IF}(\sigma < -0.999)\ \sigma \leftarrow -0.999.$$

This forces $\Gamma > 0$ (since $1 + (-0.999)\mu^2 > 0$ for $\mu \leq 1$), so $\hat{\chi} > 0$ and the standard solver runs. But the clamp:

1. **Destroys the physics**: the true velocity gradient in compression zones is replaced by a flat -0.999 plateau. The formal solution no longer represents the actual flow.
2. **Suppresses resonance structure**: in non-monotonic flows, a single observer-frame frequency can resonate with *multiple* comoving frequencies (multi-root resonance). The clamp eliminates this.
3. **Corrupts spectra**: lines formed in compression zones (shock-cooled material, cold dense shells) get wrong profiles because the velocity coupling is wrong.
4. **Breaks energy balance**: the wrong J_ν feeds into the rate equations, giving wrong populations and wrong temperatures.

5.3 The moment-equation and BA-matrix problems

Even if the formal solver could handle $\sigma < -1$, the moment equations contain terms like $(1 + \sigma)$ that become large and negative when $\sigma < -1$. The tridiagonal solver becomes unstable because the diagonal terms can change sign, the off-diagonal coupling terms reverse direction, and the Thomas algorithm assumes diagonal dominance. The same problem affects the BA matrix: the BA variation uses $(1 + \sigma)$ terms, so with $\sigma < -1$ the BA matrix can become singular or ill-conditioned.

5.4 Summary of the problem

The standard CMFGEN solver cannot handle non-monotonic velocity fields because:

1. The generalised opacity $\hat{\chi} = \chi + \Gamma/\Delta \log \nu$ becomes negative when $\Gamma < 0$, causing numerical instability.
2. The infrastructure clamp ($\sigma \geq -0.999$) prevents crashes but destroys the physics in compression zones.
3. The moment equations and BA matrix are unstable for $\sigma < -1$.
4. Multi-root resonance (multiple comoving frequencies coupling to one observer frequency) is not handled.

6 The non-monotonic solver (Knop, Hauschildt & Baron 2009)

6.1 Reference

The fix is a special formal solver that implements the formalism of

> Knop, S., Hauschildt, P. H., & Baron, E. 2009, "A non-monotonic velocity law for the radiative transfer in supernova atmospheres", *A&A*, 493, 1197.

In the codebase this is the routine `FG_J_CMF_NONMONO_V1`. It only changes *how J_ν is computed in the formal solution*; the rest of the CMFGEN iteration cycle (rate equations, temperature correction, NG acceleration, Sobolev escape probabilities, line opacities, atomic data input) is unchanged.

6.2 A simple analogy: the reversing river

Before the mathematics, here is an analogy that captures the whole idea.

The setup. Imagine a river flowing through a series of towns. The river is the *frequency axis*: each town sits at one frequency. Leaves floating on the river are *photons*. The current is the *velocity coupling* Γ: it carries leaves from one town to the next. The muddiness of the water is the *opacity* $\hat{\chi}$: it determines how far a leaf can travel before it sinks (is absorbed) or is replaced by a new one (is emitted).

Normal flow (monotonic expansion). In a normal river the current always flows *downstream* — from high frequency to low frequency (redshift). Each town receives its leaves from the town *upstream* (the next-higher frequency, which was already visited in the sweep). The muddiness is always positive (the current adds to it), so leaves travel a finite distance and everything is well-behaved. This is the standard CMFGEN solver.

The problem: a reversing river. Now imagine the river hits a gorge and the current *reverses* in some stretches — the water flows *upstream* there. This is a compression zone ($\Gamma < 0$). Two things go wrong:

1. **Negative muddiness.** The old formula added the current to the muddiness. When the current reverses, it *subtracts*, and the muddiness can go negative. A river with negative muddiness is nonsensical — leaves would *grow* stronger as they travel, exponentially. In the code this means $e^{+\hat{\tau}}$ instead of $e^{-\hat{\tau}}$: intensities blow up or go negative, and NaNs propagate.
2. **Wrong upstream town.** In the reversed stretch, leaves now arrive from *downstream* (the next-*lower* frequency), not from upstream. But the standard solver always asks the upstream town for its leaves — so it asks the wrong neighbour and gets the wrong answer.

The Knop fix, in river terms. Knop et al. (2009) make two changes:

1. **Muddiness is always positive.** Use the *strength* of the current (its absolute value $|\Gamma|$) to set the muddiness, never its signed value. The river is always muddy — leaves always attenuate. The muddiness no longer cares which way the current flows.
2. **Ask the right neighbour.** Keep the *signed* current in a separate quantity (the source-function correction Q). When $Q > 0$ (flowing downstream), ask the upstream town (the previous frequency, already computed). When $Q < 0$ (flowing upstream), ask the *downstream* town (the next frequency). The sign of Q tells each town which neighbour to listen to.

The catch: the downstream town hasn't spoken yet. CMFGEN visits towns in *downstream* order (high frequency to low frequency). In a reversed stretch, the town you need to ask (downstream, lower frequency) hasn't been visited *yet* in this round. So on the first pass you pretend it said nothing ($I_{\rm NEXT} = 0$) — a crude but stable guess. After each complete round (a "great iteration"), you *remember* what every town said and use that memory next time. Over several rounds the remembered answers converge to the truth. This is the Gauss–Seidel scheme across great iterations.

The dual-current bookkeeping. To avoid confusing the rest of the CMFGEN machinery (which was built for a one-way river), the solver keeps *two* current maps: `SIGMA_TRUE` (the real, reversing current — seen only by the new solver) and `SIGMA` (a fake, always-downstream current clamped to ≥ -0.999, seen by everything else). The old machinery keeps working as if the river never reverses; only the new formal solver knows the truth.

Summary of the analogy.

River analogy	CMFGEN quantity
River (towns along it)	frequency grid
Leaves on the water	photons (I_ν)
Current	velocity coupling Γ
Muddiness of the water	generalised opacity $\hat{\chi}$
Where leaves come from	source function S'
Upstream town	previous frequency ($I_{\rm PREV}$)
Downstream town	next frequency ($I_{\rm NEXT}$)
Reversed stretch	compression zone ($\Gamma < 0$)
One complete round of visits	one great iteration
Remembering last round's answers	Gauss–Seidel coupling
Two current maps	`SIGMA_TRUE` vs `SIGMA`

The key insight is that *resistance* (opacity) and *direction* (which neighbour to ask) are two different things. The standard solver bundled them together; Knop et al. separated them.

6.3 Core idea: separate opacity and source

The key insight of Knop et al. (2009) is to use the **magnitude** of Γ in the opacity (keeping it positive) while retaining the **signed** Γ in the source function:

$$\hat{\chi} = \chi \; + \; \frac{|\Gamma|}{\Delta \log \nu} \qquad \text{(always} > 0\text{)} \tag{18}$$

$$Q = \frac{\Gamma / \Delta \log \nu}{\hat{\chi}} \qquad \text{(signed, can be negative)} \tag{19}$$

$$S = \frac{\eta}{\hat{\chi}} \tag{20}$$

$$S' = S \; + \; Q \, I_{\text{upwind}} \qquad \text{(signed source)} \tag{21}$$

This ensures:

- $\hat{\chi}$ is always positive — no negative optical depths.
- The signed Q carries the *physical direction* of frequency coupling.
- The source function S' accounts for whether the flow is expanding ($Q > 0$) or compressing ($Q < 0$).

The transfer equation keeps the same form,

$$\frac{dI}{d\hat{\tau}} = I - S', \qquad \hat{\tau} = \hat{\chi} \, dz,$$

which is always well-defined. The physics of the compression is now captured by the *sign* of Q and the choice of upwind intensity, not by a negative opacity.

6.4 Sign-dependent upwind frequency

The **critical innovation** is the sign-dependent choice of upwind frequency.

In the standard (monotonic) solver, the upwind frequency is always the *previous* frequency (the higher frequency, already computed in the decreasing-frequency sweep). This is correct for expanding flow where $\Gamma > 0$: a photon at frequency ν couples to the next-higher frequency $\nu + d\nu$ (redshifted as it travels outward).

In non-monotonic flow, compressing regions ($\Gamma < 0$) *reverse* the frequency coupling direction: a photon at frequency ν couples to the *next-lower* frequency $\nu - d\nu$ (blueshifted as it travels into the compression). The upwind frequency is therefore the *next* frequency, not the previous one. The solver selects the upwind intensity based on the sign of Γ:

$$I_{\text{upwind}} \; = \; \begin{cases} I_{\text{PREV}} & \text{if } \Gamma \geq 0 \quad \text{(expanding: standard)} \\ I_{\text{NEXT}} & \text{if } \Gamma < 0 \quad \text{(compressing: Knop fix)} \end{cases} \tag{22}$$

6.5 Gauss–Seidel frequency coupling across great iterations

There is a catch: CMFGEN sweeps frequencies in *decreasing* order. At frequency ν_i, the next frequency ν_{i+1} has *not yet* been computed in the current sweep, so I_{NEXT} is not available directly.

The Knop et al. solution is a **Gauss–Seidel approach across great iterations**:

- On the *first* great iteration, $I_{\text{NEXT}} = 0$ (cold start). Compressing points get $S' = S$ (no frequency coupling) — a crude but stable approximation.
- After each great iteration, the computed intensities at each frequency and depth are **stored** for use as I_{NEXT} in the *next* great iteration.
- Over successive great iterations, I_{NEXT} converges to the correct value, and the frequency coupling in compression zones becomes physically accurate.

This is directly analogous to how Λ iteration converges: each iteration uses the previous iteration's radiation field, refining it until convergence.

6.6 The dual-σ architecture

To preserve compatibility with the existing CMFGEN infrastructure, the solver maintains *two* σ arrays:

Array	Value	Used by
`SIGMA_TRUE`	unclamped $d\ln V/d\ln R - 1$	the non-monotonic formal solver only
`SIGMA`	clamped (≥ -0.999)	all other routines: moment equations, BA matrix, NG acceleration, Sobolev escape probabilities, regridding, grey opacity

This separation ensures:

- The formal solution sees the *true* non-monotonic physics.
- The infrastructure routines remain *stable*.
- Existing models with monotonic velocity are completely unaffected — if there are no compressing depths, the solver behaves identically to the standard monotonic path.

`SIGMA_TRUE` is set wherever the velocity law is established (the hydrodynamical-model setup, the SN IIn velocity-law routine, the OBS module, and on warm restart), and is extended to the ray grid by linear interpolation (not in log space, because $\log(\sigma + 1)$ is undefined for $\sigma < -1$).

6.7 Bypassing the moment equations

Because the moment equations are unstable for $\sigma < -1$, the non-monotonic solver must *bypass* them. This is controlled by the VADAT flag `USE_LAM_ES`:

- `USE_LAM_ES = F`: solve the moment equations (standard path, unstable for $\sigma < -1$).
- `USE_LAM_ES = T`: use the ray-based J directly, and compute Eddington factors from the ray moments via `GET_MOMS_NON_REL` (the non-monotonic path).

The Eddington factors (K/J, N/J) computed from the ray moments are still physically correct — they come from the true formal solution — they just bypass the unstable tridiagonal moment solve.

6.8 The BA approximation in non-monotonic zones

The BA variation (`DO_VAR_CONT` → `VAR_MOM_J_CMF_V12`) uses the *clamped* σ (≥ -0.999), not `SIGMA_TRUE`. This means the BA Jacobian is *approximate* in compression zones — it does not capture the full non-monotonic physics. This is a deliberate trade-off:

- The formal solution (J computation) uses the true physics.
- The BA matrix (Jacobian) uses a clamped-σ approximation.
- With `DIAG` (`NUM_BNDS=1`), the BA iteration is stable but may converge more slowly in strongly non-monotonic regions.
- With `TRIDIAG` (`NUM_BNDS=3`), the BA matrix can go *singular* (see §9 and the "Required configuration" section below).

The Eddington factors used by the BA variation *are* computed from the true formal solution (via `GET_MOMS_NON_REL`), so the BA operator has the correct radiation-field geometry even though its velocity coupling is approximate. In practice, the iteration strategy compensates by alternating LAMBDA iterations (stable, no BA), `FULL(FIXT)` iterations (BA with approximate Jacobian), and NG acceleration (extrapolation).

6.9 Required configuration

The non-monotonic solver is activated by setting `NON_MONO_SOLVER=T` in the VADAT file. When this flag is present, the `enforce_sn_iin_physics()` routine in `prepare_VADAT.py` automatically forces a set of *non-negotiable* settings that are required for the Knop formal solver to function correctly. These forced settings cannot be overridden by the user or by the profile defaults, because incorrect values cause crashes or physically inconsistent results:

Setting	Forced value	Reason
`NON_MONO_SOLVER`	`T`	Enable the Knop et al. (2009) formal solver
`GLOBAL_LINE`	`BLANK`	Blanketing mode (lines in continuum frequency set)
`DO_SOB_LINES`	`F`	BLANK handles lines; no separate Sobolev treatment
`LAM_SET`	`F`	Do not switch to SOB for long wavelengths
`PURE_HUB`	`F`	Preserve non-monotonic velocity from STELLA
`VEL_LAW`	`11`	Read SN hydrodynamic structure directly
`USE_J_REL`	`F`	No relativistic moment solver
`NEG_OPAC_OPT`	`SRCE_CHK`	Handle negative opacity via source check
`DIF`	`T`	Diffusion inner boundary (for $\tau_{\rm inner} \geq 50$)
`IB_METH`	`DIFFUSION`	Consistent with `DIF=T`
`TRANS_*`	`BLANK`	All per-ion transitions through blanketing set

The following settings are *not* forced by the non-monotonic enforcement block, because they are iteration-strategy choices rather than solver-consistency requirements. They are set by the `single_final` profile in `prepare_VADAT.py` and can be overridden via `vadat_overrides` in the YAML config:

Setting	Current value	Reason
USE_LAM_ES	T	Bypass moment equations (unstable for $\sigma < -1$)
N_TYPE	N_ON_J	Stable at velocity reversals ($H = 0$, $J > 0$)
SOL_METH	DIAG	Depth-independent BA (stable for $\sigma < -1$)
NUM_BNDS	1	Matches DIAG
LTE_EST	T	LTE initial populations for stable cold start
FIX_T	F	T solved self-consistently (Dessart+2015)
FIX_T_AUTO	T	Auto-release FIX_T once populations stabilise
FIX_NE	F	Ne free: self-consistent NLTE ionisation
FIX_IMP	T	Freeze trace species in BA (prevents singular columns)
DO_LAM_IT	T	Start in Lambda mode (stable cold-start from LTE)
DO_LAM_AUTO	T	Auto-switch Lambda→BA when MAXCH $< 50\%$
DO_NG	T	Ng acceleration enabled (needed for DIAG convergence)
IBEG_NG	50	At least 50 iterations before first Ng
BW_NG	60	Ng bandwidth for DIAG
BEG_NG	0.1	Start Ng when MAXCH $< 10\%$
ITS/NG	8	Iterations between Ng accelerations
MAX_LAM	10^5	Max fractional pop. change per Lambda iteration
MAX_LIN	10	Max linearisation steps per BA iteration
BA_CHK_FAC	0.05	Hold BA matrix fixed when MAXCH $< 5\%$
VTURB	20 km/s	Turbulent velocity (Dessart+2015)
COH_ES	T	Coherent electron scattering for convergence
METHOD	LOGMON	Log-monotonic interpolation (Stockholm M15 ref)
INC_SHOCK_POWER	F	Shock heating already in hydro $T(r)$
THK_CONT	T	Thick continuum BC (Stockholm M15 ref)
SCL_SL_OPAC	T	Scale SL line opacities (prevent heating spikes)
REV_RGRID	T	Runtime tau-based R-grid refinement

Why DIAG, not TRIDIAG. The standard CMFGEN recommendation for monotonic flows is SOL_METH=TRIDIAG (NUM_BNDS=3), which couples adjacent depth points in the BA matrix and gives faster convergence. However, **TRIDIAG is unstable with the non-monotonic solver for strongly reversing flows**.

The root cause (documented in the BA iteration audit, BA_AUDIT_REPORT.txt) is a mathematical inconsistency in the BA variation operator:

- VAR_MOM_J_CMF_V12 (the BA variation) uses the *clamped* $\sigma \geq -0.999$ — i.e. it assumes monotonic flow in its moment-equation assembly.
- But the Eddington factors (K/J, N/J) fed into the BA operator come from GET_MOMS_NON_REL, which computes them from the *non-monotonic* formal solve (FG_J_CMF_NONMONO_V1) — i.e. they reflect the true reversing flow.
- The Eddington factors say "the radiation field has this (non-monotonic) geometry"; the moment equations say "the velocity coupling works this (monotonic) way". The resulting operator is inconsistent and can be **singular or ill-conditioned**, causing the tridiagonal solve (CMF_BLKBAND_V3) to return NaN.
- The NaN guard in SOLVEBA_V13 silently replaces NaN corrections with 0.0, so the BA iteration appears to "converge" (zero changes) while actually having *no effect*.

DIAG (NUM_BNDS=1) avoids this by solving each depth independently — there is no cross-depth coupling matrix to go singular. The trade-off is slower convergence: without the BA's global

J_ν-to-population coupling, convergence is geometric and relies on Ng acceleration to finish. The production configuration (`depth_independent_solver=True` in the YAML config, which is the default) therefore uses `DIAG`.

When TRIDIAG may work. For *mildly* non-monotonic flows (few compressing depths, σ only slightly below -1), the inconsistency is small and TRIDIAG may remain well-conditioned. The Stockholm reference models (M15-He-O-Ca-evo) converge with TRIDIAG/`N_ON_J`/`NUM_BNDS=3` in $\sim$40 iterations. But for strongly interacting SNe with deep velocity reversals (the SN2024pba case), TRIDIAG produces a singular BA matrix and must not be used. The `depth_independent_solver` flag in the YAML config controls this: `true` (default) $\rightarrow$ DIAG; `false` $\rightarrow$ TRIDIAG (experimental, for mild cases only).

Temperature handling. The current production configuration uses `FIX_T=F` with `FIX_T_AUTO=T`, following the Dessart et al. (2015) approach. This means temperature is *not* permanently fixed to the hydrodynamic value; instead, it is initially held fixed (via the `FIX_T_AUTO` bootstrap) while the populations converge from their LTE initial guess, and then automatically released so CMFGEN can solve for temperature self-consistently in the BA matrix. This prevents temperature oscillations during the cold-start phase while allowing the final model to have a physically consistent thermal structure. The earlier configuration (`FIX_T=T` permanently) is still available via `vadat_overrides` for diagnostic runs where the hydrodynamic temperature must be preserved exactly.

Electron density and impurities. `FIX_NE=F` allows CMFGEN to solve for the electron density self-consistently, which is important in the optically thin CSM where STELLA's Saha electron density may be inaccurate. `FIX_IMP=T` freezes trace species (low-abundance ions) in the BA matrix via `FIXPOP_IN_BA_V3`, preventing singular BA matrix columns that would otherwise arise from trace species with near-zero populations.

Ng acceleration. Ng acceleration is *enabled* (`DO_NG=T`) in the current configuration, with `IBEG_NG=50` (at least 50 iterations before the first Ng) and `BW_NG=60` (bandwidth). This is essential because the DIAG solver converges geometrically — without Ng, pure Lambda iteration stalls at $\sim 10^{10}$ after 100+ iterations. The Ng implementation in `do_ng_accel_band_v4.f` includes robust safety mechanisms: per-depth rejection if any population changes by more than $10.1\times$ or less than $0.09\times$, global cancellation if more than 3 depths fail, and a post-Ng Lambda fallback if corrections exceed 50%. Ng only fires when `MAXCH` $<$ `BEG_NG` (10%) and `MAIN_COUNTER` $\geq$ `IBEG_NG` (50), by which point BA mode is active and `FIXPOP_IN_BA_V3` freezes trace species.

The DIFFERENCE method (`FG_OPT=DIFF`) is rejected because it does not construct $\hat{\chi} = \chi + |\Gamma|/\Delta \log \nu$ and is unstable for $\sigma < -1$. `N_ON_J` is required because at velocity reversals the first moment H passes through zero, making N/H (`G_ONLY`) singular; N/J is always well-defined since $J > 0$.

6.10 NaN safety guards

Even with the correct integrals, numerical noise can produce slightly negative intensities or (worse) NaNs, which on the next iteration would propagate through the whole grid. The solver therefore includes comprehensive NaN guards at five points in the propagation chain:

1. **Ray intensity clamping** ($I_M, I_P \geq 10^{-30}$ if < 0 or NaN) — catches NaN from source-function advection in compression zones. Uses the IEEE-754 self-comparison trick ($x \neq x$ is true only for NaN) because Fortran's $x < 0$ test returns `FALSE` for NaN.
2. `JNU_STORE` **clamping** ($\geq 10^{-30}$ if $< 10^{-30}$ or NaN) — prevents division by zero in $f_{\rm Edd} = K/J$ and NaN Eddington factors entering the BA matrix.
3. `I_NEXT` **storage guard** (NaN check before writing to the Gauss–Seidel arrays) — prevents NaN contaminating the next great iteration.

4. **TC/FEDD clamping in** `COMP_J_BLANK` (after the formal solve, before the moment solver) — TC $\geq 10^{-30}$, FEDD ≥ 0 with default 1/3 if NaN.
5. K/J **division guard in** `GET_MOMS_NON_REL` — clamps `JNU_STORE` and `KNU_STORE` before computing K/J, preventing NaN K/J from entering the BA matrix.

All guards use OpenMP `ATOMIC` operations on the diagnostic counters to avoid race conditions in the parallel ray loop.

6.11 Sparse storage optimisation

Storing intensities at *all* depths, all rays, and all frequencies for the Gauss–Seidel $I_{\rm NEXT}$ arrays would require enormous memory: $N_D \times N_P \times N_{\rm CF} \times 8$ bytes per array, which can reach tens of GB for realistic grids.

The optimisation exploits the fact that only depths where $\sigma_{\rm true} < -1$ can ever have $\Gamma < 0$ (for any ray angle). These are the "compressing" depths. The solver:

1. Counts $N_{\rm comp}$ = number of depths with $\sigma_{\rm true} < -1$.
2. Builds a sparse index `COMP_DEPTH_IDX` mapping the full-depth index to a sparse index $1 \ldots N_{\rm comp}$ (or 0 if not compressing).
3. Allocates `I_P_DEPTH_NEXT` and `I_M_DEPTH_NEXT` with dimensions $(N_{\rm comp}, N_P, N_{\rm CF})$ — typically much smaller than $(N_D, N_P, N_{\rm CF})$.

If there are no compressing depths (`HAS_COMPRESSING = .FALSE.`), the large arrays are not allocated at all, and the solver behaves identically to the standard monotonic path. An additional in-place optimisation writes intensities directly to the `NEXT` arrays during the frequency sweep, eliminating the need for separate current-iteration arrays and the initial copy/zero.

6.12 Warm restart handling

On warm restart (`NEWMOD = .FALSE.`), the routine that normally sets `SIGMA_TRUE` is not called, so `SIGMA_TRUE` may be zero from allocation. The solver re-derives it from V and R,

$$\sigma_{\rm true}(I) \;=\; \frac{R(I)}{V(I)}\,\frac{dV}{dR} \;-\; 1,$$

using centred differences for interior points and one-sided differences at the boundaries. In the OBS module, `SIGMA_TRUE` is always re-derived from V and R because the restart file stores only the clamped σ.

6.13 Diagnostics

On the first frequency of the first iteration the solver prints:

- the number of compressing depths ($N_{\rm comp}$);
- the number of depths with $\sigma < -1$;
- the $\sigma_{\rm true}$ range (min, max);
- the number of rays with multi-root resonances;
- the resonance turning-point count (min, max);
- confirmation of $\hat{\chi}$, signed Q, and sign-dependent upwind.

During the run it tracks (for an end-of-run summary) the count of clamped negative/NaN intensities and the count of clamped NaN mean intensities.

7 Putting it together: what actually happens in a run

1. A hydrodynamics code (e.g. STELLA) gives the structure — density, velocity, temperature, composition vs radius — at a chosen snapshot time.
2. This is converted into `SN_HYDRO_DATA` (CMFGEN's input format, indexed outside-in: depth 1 = outermost shell).
3. CMFGEN builds an atomic model (which ions, which levels) based on the physical conditions.
4. It iterates: Λ iterations to get close, then BA/Full iterations to converge. At every iteration the non-monotonic solver handles the compression zones so J_ν stays physical.
5. When the populations and luminosity stop changing, CMFGEN does a final formal solution and writes out the emergent spectrum via the OBS module.

8 The STELLA-to-CMFGEN conversion pipeline

The conversion from a STELLA snapshot to a CMFGEN `SN_HYDRO_DATA` file is performed by `convert_stella_swd_to_cmfgen.py` and is a critical step that determines the quality of the CMFGEN run. The current pipeline follows the Dessart et al. (2015) approach with several project-specific adaptations.

8.1 Tau-based truncation (Dessart+2015)

The STELLA grid is first truncated to the region $10^{-5} \leq \tau_{\rm Rosseland} \leq 30$, following Dessart et al. (2015). The outer boundary is set by the NLTE optical depth ($\tau_{\rm min} = 0.01$), and the inner boundary by the constant-kappa Rosseland optical depth with $\kappa = 0.34$ cm^2 g^{-1} (typical SN ejecta opacity). The constant kappa is used for the inner boundary because STELLA's NLTE kappa is too low in cold neutral gas (giving $\tau_{\rm max} \sim 1$ instead of ~ 240 with $\kappa = 0.34$), so $\tau_{\rm max} = 30$ would never be enforced with the NLTE kappa. The truncation is *strict*: if too few points remain, the grid is not relaxed — the adaptive grid algorithm adds points where gradients require them, but the $\tau = 30$ boundary is preserved.

8.2 Adaptive grid refinement (AMR)

The truncated STELLA grid is remapped onto a CMFGEN depth grid using a tau-based adaptive mesh refinement algorithm. The algorithm:

1. Starts from the tau-trimmed STELLA grid as the base (typically 80–150 points after truncation).
2. Computes diagnostic gradients: radius, temperature, density, electron density, velocity, optical depth, and the velocity structure parameter σ.
3. Identifies the worst-violating edge (highest score relative to its threshold) and inserts a midpoint.
4. Repeats until all criteria are satisfied or `max_points` is reached.
5. Inserts `n_outer_extra`=5 additional points in the outermost low-tau region for better spectral accuracy.
6. Runs a final grid+sigma smoothing pass to resolve any remaining violations.

The current AMR criteria and thresholds are:

Criterion	Threshold	Description
`max_radius_jump`	0.05	Max fractional radius jump dr/r
`max_temp_jump`	0.10	Max fractional temperature jump dT/T
`max_density_jump`	0.15	Max fractional density jump $d\rho/\rho$
`max_ne_jump`	0.10	Max fractional electron density jump
`max_logtau_jump`	0.10	Max $\Delta \log \tau$ per cell
`max_velocity_jump`	0.10	Max fractional velocity jump dV/V
`max_sigma_jump`	1.0	Max $\lvert\Delta\sigma\rvert$ per cell
`max_cell_ratio`	1.5	Max ratio of adjacent cell sizes
`max_points`	100–150	Upper grid size limit
`min_points`	60–80	Minimum grid size (fill if below)

Ionisation fronts are detected separately (via $|d \log n_e / d \log r| > 0.5$) and force-include a minimum of 5–10 source zones around each front to ensure adequate resolution. The AMR algorithm uses a "worst-first" insertion strategy: the single worst-violating edge is refined at each step, guaranteeing that the highest-gradient regions receive points first.

8.3 Velocity smoothing

Small numerical oscillations in the STELLA velocity profile ($|\Delta V|/V \sim 1\%$ over a few zones) are smoothed with a Savitzky–Golay filter (window 5 zones, order 2) to prevent the structure parameter σ from reaching unphysical values ($\sigma \ll -100$) that would destabilise the CMFGEN Sobolev integrals. The smoothing preserves the peaks of the velocity profile (unlike Gaussian smoothing, which would reduce the shock velocity jump) by redistributing radius rather than convolving values. Velocity smoothing can be disabled via the YAML config (`smooth_velocity: false`) for models where the STELLA velocity is already smooth.

8.4 Temperature profile

The temperature profile from STELLA is used directly, with a floor of $T_{\rm floor} = 2000$–5000 K (configurable) to prevent numerical issues in the cold outer CSM. The Dessart et al. (2015) adiabatic temperature profile is preserved in the conversion: the STELLA gas temperature is used without modification, and the CSM temperature is set to 6000 K (typical photoionised CSM temperature) where the STELLA profile drops below the floor.

8.5 Composition and Ni56

The STELLA 15-element composition is mapped onto CMFGEN's species list. The ^{56}Ni distribution is read from the STELLA `.xni` file (Fortran unformatted `REAL*8`) and included in the `SN_HYDRO_DATA` abundance table so that CMFGEN can compute radioactive decay heating self-consistently. For interaction-only models with no ^{56}Ni (the SN_B case), a zero `.xni` file is generated and the slurm template zeroes the `.xni` file after the `eve2` pre-processor runs, ensuring no spurious radioactive decay.

9 Caveats and limitations of the non-monotonic solver

The Knop et al. (2009) solver makes the formal solution stable and physically meaningful in compression zones, but it is *not* a complete non-monotonic radiative-transfer code. This section lists

the known caveats so that users can judge when the results are reliable and when a different tool (e.g. a ray-by-ray Lagrangian solver, or a code such as SEDONA/SuperLite that does not assume monotonic flow) is more appropriate.

9.1 It is still a Sobolev / single-resonance approximation

The solver operates within CMFGEN's comoving-frame formal-solution framework, which inherits the Sobolev (large-velocity-gradient) picture: each frequency is assumed to couple to *one* upwind frequency at each depth. In a genuine velocity *reversal* a single observer-frame frequency can resonate with *multiple* comoving frequencies along a single ray (multi-root resonance). The solver picks a single upwind frequency based on the *local* sign of Γ; it does not trace all resonance points. Lines formed deep inside a strongly reversing flow can therefore have wrong profiles even though J_ν stays finite.

Implication: reliable for shocks and decelerating shells where the reversal is mild or where the line forms outside the reversal; unreliable for lines that form *inside* a deep velocity minimum with multiple resonance points.

9.2 The σ clamp is numerical, not physical

Every CMFGEN routine *except* the non-monotonic formal solver sees the clamped $\sigma \geq -0.999$. This includes:

- the moment equations (`MOM_J_CMF_V11`, `MOM_JREL_V9`);
- the BA / variation matrix (`VAR_MOM_J_CMF_V12`, `VAR_JREL_V5`);
- the Sobolev escape-probability routines (`SOBEW`, `SOBJBAR`, `BETANEW`);
- the grey opacity solve (`JGREY_WITH_FVT`);
- NG acceleration and regridding.

The clamp creates a flat $\sigma = -0.999$ plateau wherever the true $\sigma < -1$. In those plateau regions the moment equations and the BA Jacobian are *approximate*: they use a well-conditioned but unphysical velocity gradient. The formal solution itself sees the true σ (via `SIGMA_TRUE`), so J_ν is correct, but the *coupling* between J_ν and the population correction (the BA step) is approximate in compression zones.

Implication: convergence in strongly non-monotonic regions may be slower than in monotonic regions, because the BA operator is not the true Jacobian there. The iteration strategy (alternating LAMBDA, FULL(FIXT), and NG) compensates, but a model with a large fraction of compressing depths may need many more iterations.

9.3 The BA matrix uses an approximate Jacobian in compression zones

As noted above, the BA variation (`DO_VAR_CONT` $\rightarrow$ `VAR_MOM_J_CMF_V12`) uses the clamped σ, not `SIGMA_TRUE`. This is a deliberate trade-off: the true $\sigma < -1$ would make the BA matrix singular or ill-conditioned (because the $(1 + \sigma)$ terms change sign), so the clamped value is used to keep the linear solve well-posed. The Eddington factors fed into the BA operator *are* computed from the true formal solution (via `GET_MOMS_NON_REL`), so the radiation-field geometry is correct even though the velocity-coupling part of the Jacobian is approximate.

With DIAG (`NUM_BNDS=1`, the production setting): the BA step is stable. Each depth is solved independently, so there is no cross-depth coupling matrix to go singular. The BA correction is approximate in compression zones (the Jacobian uses clamped σ), so convergence there may be slower than in monotonic flow. In pathological cases the BA correction can be too small, requiring extra LAMBDA or NG iterations to finish.

With TRIDIAG (`NUM_BNDS=3`, experimental): the BA matrix *can go singular* for strongly non-monotonic flows. The cross-depth coupling in the tridiagonal system combines the clamped σ (monotonic moment equations) with non-monotonic Eddington factors, producing an inconsistent operator. In the SN2024pba tests this caused `CMF_BLKBAND_V3` to return NaN, which the `SOLVEBA_V13` NaN guard silently replaced with 0.0 — making the BA iteration have zero effect while appearing to converge. This is why the production configuration uses DIAG. TRIDIAG may work for mildly non-monotonic flows (see the Stockholm M15 reference) but must not be used for deep velocity reversals.

9.4 Gauss–Seidel coupling converges slowly on the first great iterations

The sign-dependent upwind frequency in compression zones requires $I_{\rm NEXT}$, which is not yet computed in the current decreasing-frequency sweep. The solver uses a Gauss–Seidel scheme *across great iterations*: on the first great iteration $I_{\rm NEXT} = 0$, so compression zones get $S' = S$ (no frequency coupling). The coupling is filled in only over successive great iterations.

Implication: the *first* great iteration of a non-monotonic model is a cruder approximation than a monotonic model's first iteration. If the model is restarted from a cold start (LTE populations) with strong compression, the early iterations may show large population swings before the Gauss–Seidel coupling converges. A warm restart from a monotonic model of the same snapshot reduces this problem.

9.5 Velocity plateaus and discontinuities are not handled well

The Sobolev approximation assumes the velocity scale height is much larger than the photon mean free path. Two regimes violate this:

- **Velocity plateaus** ($dV/dR \approx 0$): $\sigma \to -1$, $\Gamma \to 0$, and the generalised opacity reduces to the static opacity. The frequency coupling vanishes, so the comoving-frame advantage is lost and the line profile is unresolved.
- **Sharp discontinuities** (true shocks with $|\Delta V|/V \sim 1$ over one zone): the centred-difference σ is dominated by interpolation artefacts, and the single-resonance assumption fails because the resonance point jumps discontinuously.

CMFGEN's grid should resolve shocks over several zones; the `max_sigma_jump` AMR criterion in the conversion pipeline is designed for this. A shock that fits in a single grid cell will produce unreliable line profiles regardless of the solver.

9.6 Flux conservation has not been quantitatively verified

The solver preserves J_ν positivity and finiteness, and the Eddington factors come from the true formal solution, but a *quantitative* flux-conservation test (comparing the integrated emergent luminosity to the input luminosity + radioactive + interaction deposition) has not been performed for the non-monotonic path. The algebraic analysis shows that the Knop et al. formulation is conservative, but numerical round-off, the σ clamp, and the Gauss–Seidel lag can introduce small flux errors in compression zones.

Implication: treat the absolute luminosity of lines formed in compression zones as uncertain at the ∼few-percent level. The *shape* of the spectrum and the relative strengths of lines formed outside compression zones are reliable.

9.7 Restart reproducibility is not guaranteed

On warm restart (`NEWMOD = .FALSE.`), `SIGMA_TRUE` is re-derived from V and R by finite differences, because the restart file stores only the clamped σ. The re-derivation uses centred differences on the *current* grid, which may differ slightly from the grid on which the original σ was computed (if regridding occurred). This means a restarted run may not be bit-for-bit reproducible with respect to the original run in the compression zones.

9.8 Very small grids ($N_D < 100$)

The non-monotonic solver inserts boundary points and requires enough interior points to resolve the compression region with centred differences. On very small grids ($N_D < 100$) the boundary insertion can leave too few interior points, the σ finite differences become noisy, and the formal solution can oscillate. The AMR pipeline targets $N_D \sim 150$–300 for interacting-SN models, which is safe.

9.9 Extreme $\sigma \ll -1$

When σ is very negative (e.g. $\sigma = -7.5$ at a strong reverse shock), the clamp plateau is wide and the discrepancy between the true physics and the clamped BA Jacobian is large. The formal solution is still correct (it uses `SIGMA_TRUE`), but the BA iteration may stall or converge to a slightly wrong population distribution in those zones. For such extreme cases, a Monte Carlo code (SEDONA, SuperLite) that does not rely on the Sobolev approximation is a better cross-check.

9.10 What is *not* changed by the solver

It is worth restating what the non-monotonic solver does *not* touch, so that users know where to look for unrelated problems:

- the rate equations (`SOLVE_FOR_POPS`);
- the atomic data input and level solving;
- the temperature correction (`DIFFW` / radiative equilibrium);
- NG acceleration (uses clamped σ, unchanged);
- Sobolev escape probabilities (Dessart+2015 guard, separate from the Knop solver);
- line opacities and continuum opacities;
- the OBS module (spectrum post-processing).

A convergence problem or a wrong line ratio that persists with the non-monotonic solver off is *not* caused by the non-monotonic solver and should be diagnosed with the standard CMFGEN toolkit.

9.11 When to use a different code

The non-monotonic solver is the right tool when:

- the velocity field is mostly monotonic with a mild, resolved compression zone (shock, decelerating shell);
- the scientifically important lines form *outside* the compression zone;
- a detailed non-LTE spectrum with many explicit levels is needed (CMFGEN's strength).

A different tool is preferable when:

- the flow has a deep, multi-resonance velocity reversal and the lines of interest form inside it — use SEDONA or SuperLite (Monte Carlo / Lagrangian, no Sobolev assumption);

- the shock is unresolved (< 3 zones) — refine the grid or use a code with shock-capturing hydrodynamics coupled to the transport;
- time-dependent radiation transport matters (e.g. shock breakout) — CMFGEN is a snapshot code; use STELLA or a time-dependent RHD code.

10 Key takeaways

- CMFGEN solves *coupled* radiative transfer + statistical equilibrium by *iteration*, accelerated with NG and BA methods.
- Almost all of its machinery assumes the velocity increases monotonically outward (Sobolev picture), so the generalised opacity $\hat{\chi} = \chi + \Gamma/\Delta \log \nu$ is always positive.
- Interacting supernovae have *compression zones* where $dV/dR < 0$, breaking that assumption and making $\sigma < -1$, so Γ can be negative and $\hat{\chi}$ can go negative.
- The infrastructure clamp ($\sigma \geq -0.999$) prevents crashes but destroys the physics in compression zones.
- The Knop et al. (2009) non-monotonic solver fixes this by:
  1. using $|\Gamma|$ in the opacity (so $\hat{\chi} > 0$ always);
  2. retaining the signed Γ in the source function (so the physical direction of frequency coupling is preserved);
  3. choosing the upwind frequency from the sign of Γ (previous frequency for expanding flow, next frequency for compressing flow);
  4. using a Gauss–Seidel scheme across great iterations to supply the not-yet-computed $I_{\rm NEXT}$ in compression zones;
  5. maintaining a dual-σ architecture so the formal solver sees the true physics while the infrastructure routines stay stable;
  6. bypassing the unstable moment equations (`USE_LAM_ES = T`) and computing Eddington factors directly from the ray moments;
  7. using the clamped σ in the BA matrix with `DIAG` (`NUM_BNDS=1`) for stability — the depth-independent solve avoids the singular cross-depth coupling that TRIDIAG produces when non-monotonic Eddington factors meet clamped-σ moment equations;
  8. guarding against NaNs and negative intensities at five points in the propagation chain.
- It only needs to run on the compression depths, so a sparse storage optimisation keeps memory manageable.
- The rest of CMFGEN (rate equations, temperature correction, NG acceleration, Sobolev probabilities, line opacities, atomic data) is unchanged — the solver is a drop-in replacement for the formal-solution step.
- The current production configuration (SN2024pba) uses:
  - LTE cold start (`LTE_EST=T`) for stable initial populations;
  - pure Lambda iterations with auto-switch to BA (`DO_LAM_IT=T`, `DO_LAM_AUTO=T`);
  - Ng acceleration enabled (`DO_NG=T`, `IBEG_NG=50`, `BW_NG=60`) — essential for DIAG convergence;
  - temperature solved self-consistently with bootstrap (`FIX_T=F`, `FIX_T_AUTO=T`, Dessart+2015);
  - electron density free (`FIX_NE=F`) for self-consistent NLTE ionisation;
  - trace species frozen in BA (`FIX_IMP=T`) to prevent singular matrix columns;
  - turbulent velocity 20 km/s (`VTURB=20`, Dessart+2015);
  - coherent electron scattering (`COH_ES=T`) for convergence;
  - blanketing line treatment (`GLOBAL_LINE=BLANK`, forced by the non-monotonic enforce-

ment);
- tau-based AMR grid with $\tau_{\rm Rosseland} \leq 30$ truncation (Dessart+2015) and ionisation-front-aware refinement;
- diffusion inner boundary (`DIF=T`, `IB_METH=DIFFUSION`, forced by the non-monotonic enforcement).